ARTS Place du 11 Aout Cluny 71250

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Sender : **Lamice DENGUIR**
Ecole Nationale Supérieure d'Arts et Métiers
LaBoMaP (EA 3633) – Equipe UGV
ENSAM Rue Porte de Paris – Cuny 71250

Object

ALE modeling applied to orthogonal cutting of OFHC copper

\- ***by Lamice DENGUIR -***

- Corresponding project: USI COR SURF-

ALE modeling applied to orthogonal cutting of OFHC copper

1- General description

In this ALE model, Eulerian surfaces are defined in order to allow the motion of the workpiece material flux. Nodes at the bottom of the workpiece are pinned vertically ($U_y = 0$) and the tool is totally fixed. Concerning tool/chip interface, a « penalty » contact was used with master-slave surfaces. The friction at the interface used is a Coulomb friction. For heat transfer boundary conditions, only the flux between the tool and the workpiece are considered.

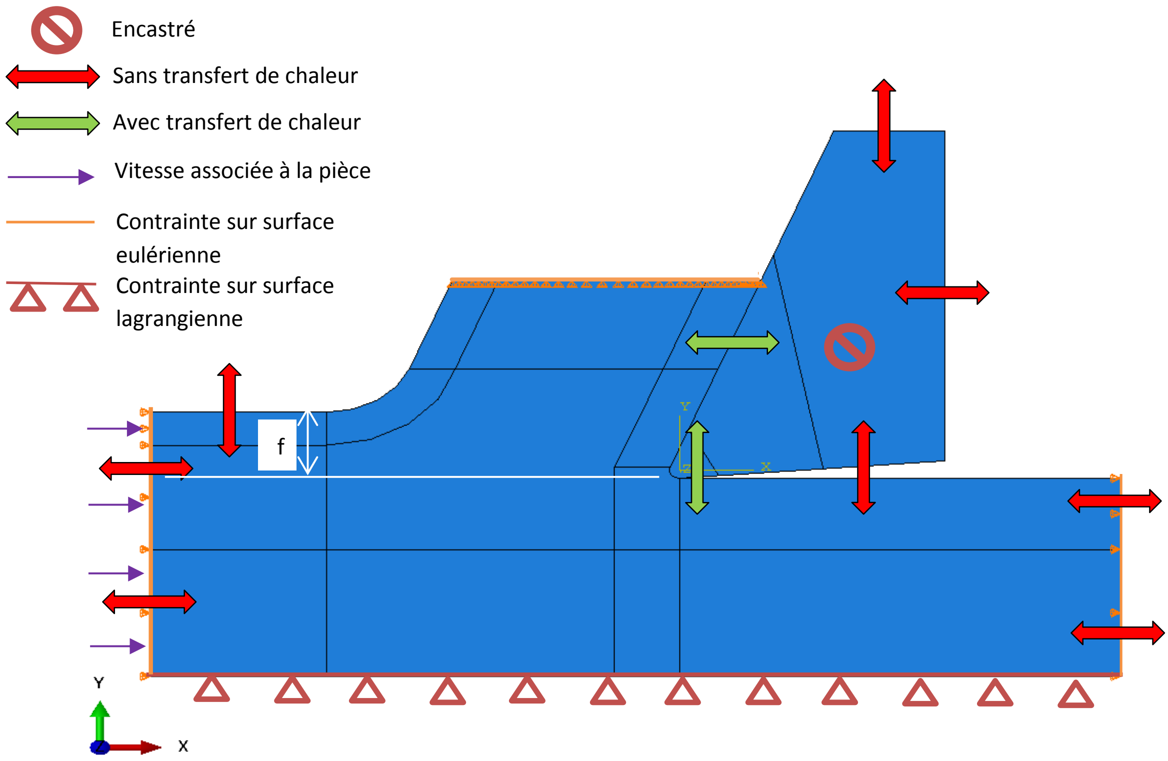

Figure 1. Boundary conditions for ALE formulation

2. Modeling on ABAQUS Explicit

The file corresponding to the following description is **ALE25_Nasr.cae** and the corresponding results are figuring in the job **h100essai76.odb**.

2.1. Tool and workpiece geometry

- Uncut chip thickness **h = 100 µm**

- Tool's edge radius $\mathbf{r_n}$ **= 12 µm**

- Tool's rake angle **γ = 10°**

- Tool's flank angle **α = 5°**

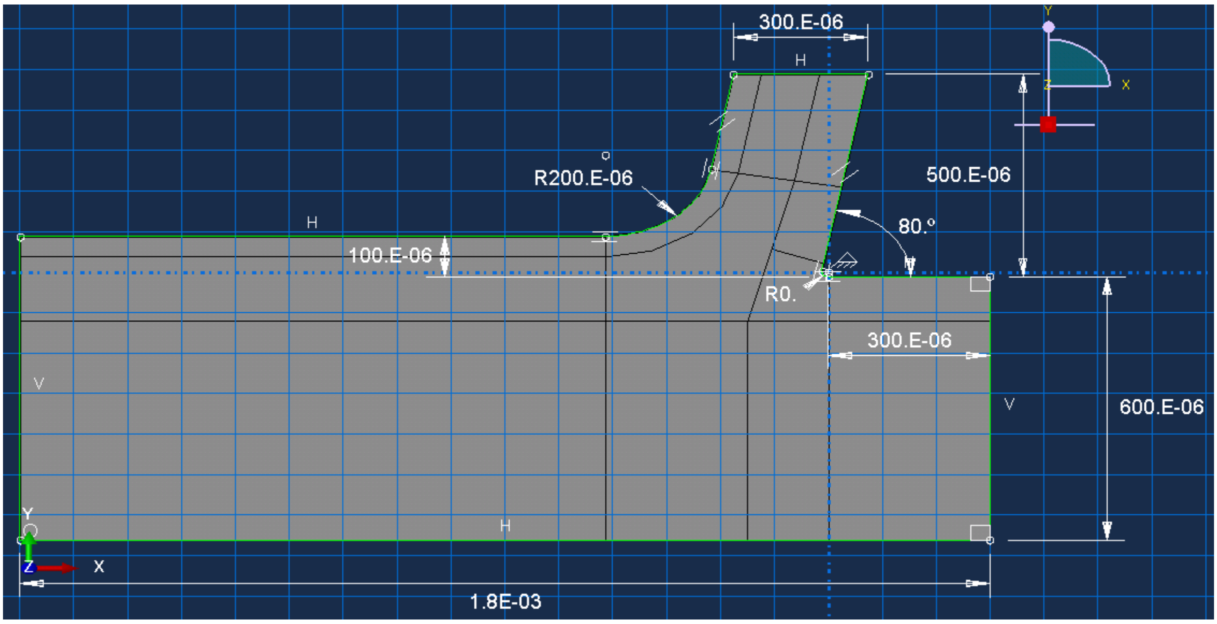

Figure 2. The workpiece sketch

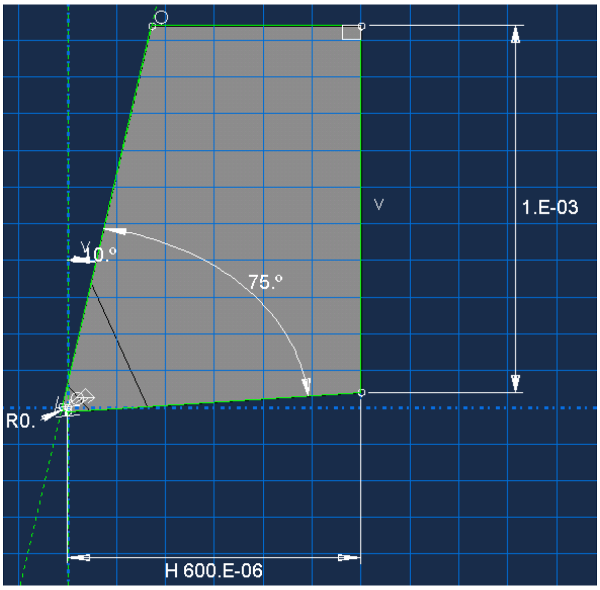

Figure 3. The tool geometry

2.2. Materials properties

<u>a) For the workpiece : OFHC copper-JC</u>

	Unit	Value for OFHC copper
General		
Density	kg/m^3	8940
Mechanical properties		
Young modulus E	GPa	118
Poisson coefficient υ		0.33
Johnson Cook parameters		
A	MPa	90
B	MPa	292

n		0.31
m		1.09
Tm	°K	1358
T0	°K	293
C		0.025
$\dot{\varepsilon}$		1
Thermal properties		
Conductivity κ	W/mK	Temperature dependent
Expansion α	$°K^{-1}$	Temperature dependant
Specific heat C_p	J/kgK	Temperature dependant
Inelastic heat fraction		0.9

Table 1. OFHC copper used properties

b) For the tool : Carbide

	Unit	Value for OFHC copper
General		
Density	kg/m^3	14450
Mechanical properties		
Young modulus E	GPa	630
Poisson coefficient υ		0.3
Thermal properties		
Conductivity κ	W/mK	Temperature dependent
Specific heat C_p	J/kgK	Temperature dependant

Table 2. Carbide used properties

2.3. Interactions

a) Interaction between the tool and the workpiece : Int-outil-massif

The mechanical constraint formulation uses the penalty contact method with finite sliding. The contact interaction property is defined by a dry contact with a Coulomb friction (**IntProp-interface-sec**, μ = 1.12)

b) Self contact : Int-self-contact

The mechanical constraint formulation uses penalty contact method. In the first step, this interaction doesn't have an influence on the results that the chip wouldn't get in contact with the workpiece surface.

2.4. Boundary conditions

a) Temperatures

- The whole system has initially a temperature of **293°K**

- The input flow material (in **Set-entree**) is supposed to have the room temperature of **293°K** (**BC-temperature**)

- The tool borders defined in (**BC-Temp tool**) are supposed to have the room temperature of **293°K.**

b) Velocities

- The input flow material has a uniformly distributed velocity of **2 m/s** in the X direction (**V_1=2**) over the eulerian surface (**surf-entree**), but in order to avoid the excessive distortion of the mesh at the beginning of the calculous, this velocity is multiplied by a smooth shaped amplitude (**Amp-sstep**) moving from 0 at t=0s to 1 at t=10 ms. For that reason, the steady state is supposed to be established after **10 ms** of the simulated time.

- The bottom of the workpiece is considered as a lagrangian surface (**set-massif-fixe**) and its motion is constrained by (**U_2=UR_1=UR_3=0**) that it is supposed to move only in the direction X of the input flow material.

- The tool is initially entirely fixed (**encastré U_1=U_2=U_3=UR_1=UR_2=UR_3=0**)

2.5. The mesh

Both the tool and the workpiece are meshed within quadrilateral elements **CPE4RT** in plain strain, temperature-displacement coupled. The mesh of the workpiece is refined near scission zones.

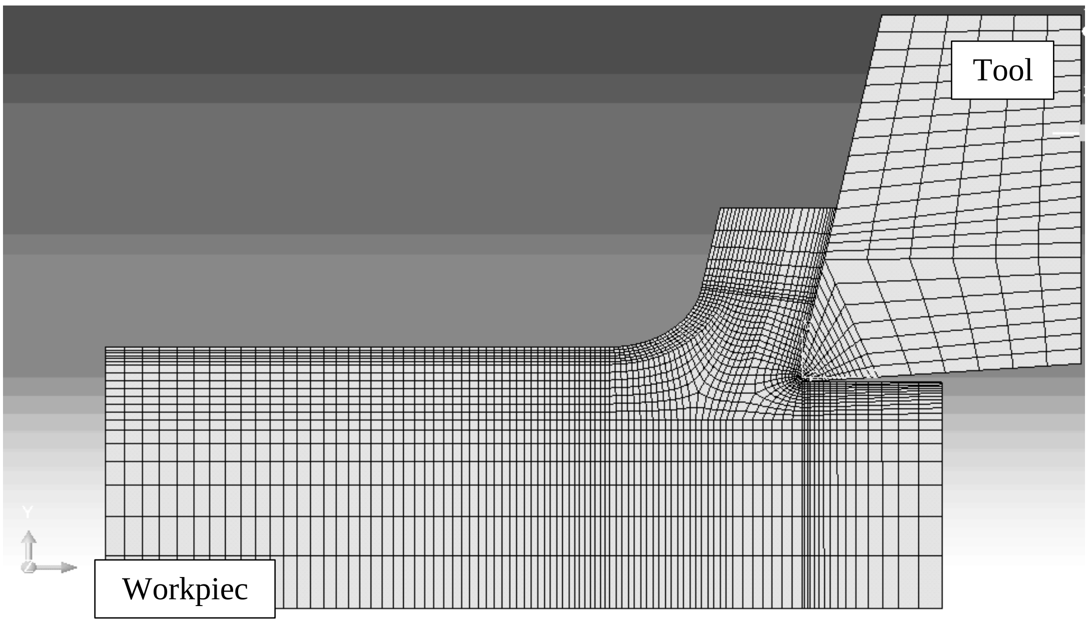

Figure 4. The assembly mesh

The approximate global size of the elements constituting the workpiece is of **10 µm**.

2.6. ALE Adaptive mesh

a) Adaptive mesh domain

- The ALE adaptive mesh domain chosen is the whole workpiece (region: **Set-massif**)

- The **remeshing sweeps per increment = 50**, and so the initial remeshing sweep.

- The frequency cannot be different from 1 that the adaptive mesh constraints are activated with a motion independent from the underlying material.

b) Adaptive mesh constraints

- The top surface of the chip is considered as an eulerian surface. The flow material is not allowed to be represented moving above it, so the adaptive mesh constraint is set in the region (**Set-copeau-fin**) with **U_2=0**.

- As the input flow material is coming from the surface (**Set-entree**), it is considered as an eulerian surface, and so this surface is not allowed to be represented moving in both directions X and Y (**U_1=0 and U_2=0**)

- As the output flow material moves through the surface (**Set-sortie**), it is not allowed to be represented moving in the X direction. (**U_1=0**)

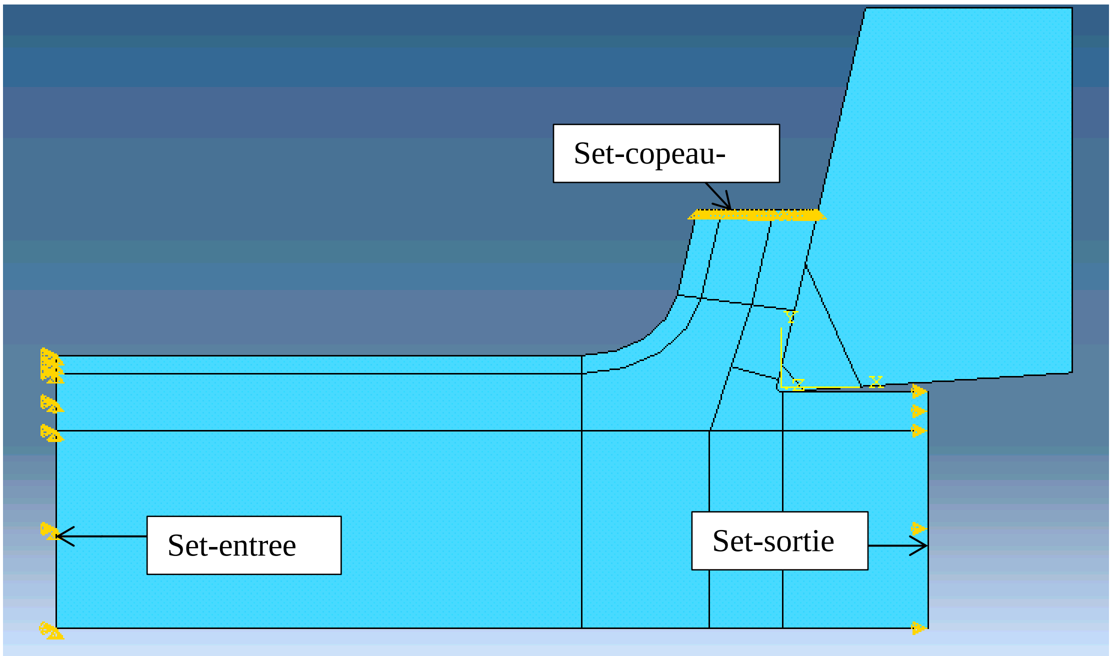

Figure 5. Adaptive mesh constraints

2.7. Step

- The chosen step type is: Dynamic, temperature-displacement coupled, explicit.

- The time period simulated is **0.02 s**.

- Mass scaling is activated with a target time increment = 3E-08s and a factor of 1E03

2.8. Job

- The job type is a full analysis using **18 processors** with **18 domains**.

- The number of the **total simulated increment is 670605** for **the total simulated time of 0.02 s**.

- The **CPU time** taken to finish the simulation is **54738.9 sec**. (submitted at 8:56 and completed at 09:05 the next day).

3. Remarks

- It is normal in an ALE model that the chip thickness initially grows significantly and then collapses again. In fact, the initial geometry of the chip is introduced by the operator, basing on the experimental result. The values reached at the steady state are then independent from the initial values.

- This transitory phenomenon has been seen in previous studies

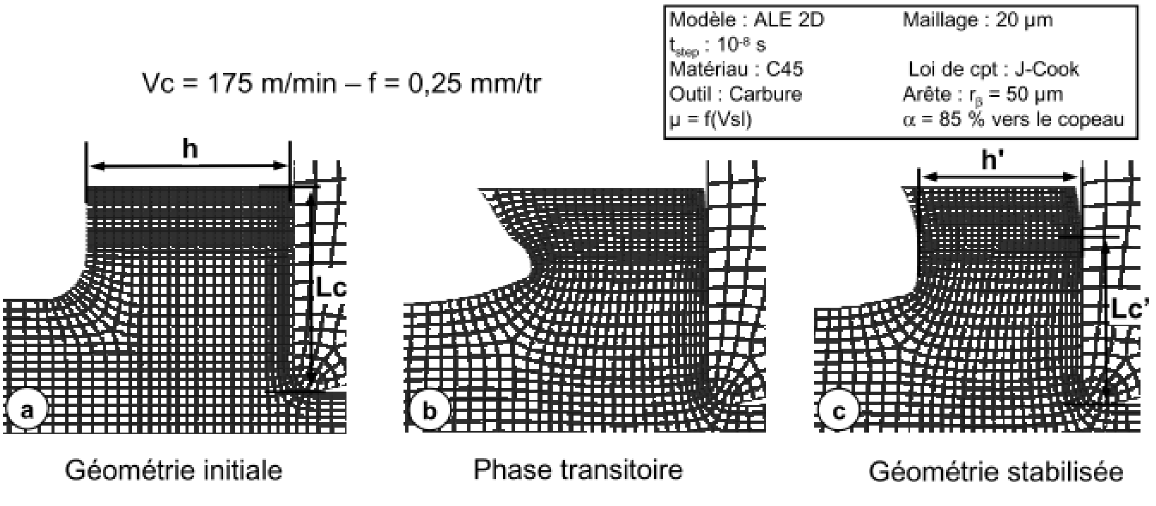

FIGURE A.2: Evolution de l'épaisseur du copeau et de la longueur de contact entre la géométrie initiale et la phase stabilisée

[C. Courbon, 2011, Vers une modélisation physique de la coupe des aciers spéciaux : intégration du comportement métallurgique des phénomènes tribologiques et thermiques aux interfaces]

- Moreover, the fact of multiplying the input velocity by smoothly shaped amplitude before reaching the steady state can amplify the modifications in the chip geometry during the transitory period

- The problem that we are facing now is that the chip compression ratio obtained in the steady state is far from what I have found experimentally. Indeed, in this model, the CCR is = 1.6, however the real one is almost = 6.